

# Hasi Hays, Ph.D.

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Born: 1985 - Sri Lanka  
Nationality - Sri Lankan

**Key areas of research:** Computational Systems Biology, Machine Learning, Artificial Intelligence  
**Experience summary:** Industrial - 6 years (Biotech), Academic - 5 years (postdoctoral), Ex-military Captain.

## Education

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| 2021      | <b>MIT xPRO</b> - Principles of biomanufacturing: Using biotechnology to manufacture medicine, Massachusetts Institute of Technology (MIT), USA.                                                         |
| 2016-2020 | <b>Ph.D.</b> Specialized in Computational Biology (1 <sup>st</sup> class honors), GPA = 4/4, Awarded the “ <b>Honor of excellent international graduate of 2020</b> ”, Ocean University of China, China. |
| 2012-2016 | <b>M.Sc.</b> in Food science and technology (2 <sup>nd</sup> upper honors), University of Peradeniya, Sri Lanka.                                                                                         |
| 2007-2011 | <b>B.Sc.</b> in Animal science special degree, (2 <sup>nd</sup> upper honors) Uva Wellassa University, Sri Lanka.                                                                                        |

## Current Positions

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|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023- | <b>Postdoctoral research associate</b> - Chemical Engineering. Research areas: Computational Systems Biology, Foundation Models, Machine Learning, Artificial Intelligence, Graph Neural Networks. CHEG Computer Modeling Lab, Dept. of Chem. Eng., University of Arkansas, USA. |
| 2021- | <b>Review editor in artificial intelligence</b> - Medicine and public health, Frontiers journal.                                                                                                                                                                                 |
| 2024- | <b>Member</b> - National postdoctoral association, USA.                                                                                                                                                                                                                          |

## Past Positions

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|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022-2023 | <b>Postdoctoral research scientist</b> - AI and big data in cancer research to identify somatic mutations and develop precision medicine, Institute of Biochemistry, Molecular Biology & Biotechnology (IBMBB), University of Colombo, Sri Lanka. |
| 2018-2023 | <b>Chief scientist</b> - Multi-omics, big data, and precision medicine, Wixsee, Sri Lanka.                                                                                                                                                        |
| 2021-2023 | <b>Foreign expert consultant</b> - Biomedical data analysis, Yantai city development, Shandong, China.                                                                                                                                            |
| 2021-2022 | <b>Chief consultant</b> (biotech) - Lysine production plant, Oryzas Organics (PVT) Ltd., Sri Lanka.                                                                                                                                               |
| 2020-2021 | <b>Director</b> - Global technical service, Qingdao GBW Group (Biotech), China.                                                                                                                                                                   |
| 2014-2016 | <b>Production officer</b> - National aquaculture group (NAQUA), Saudi Arabia.                                                                                                                                                                     |
| 2014      | <b>Captain</b> - Direct enlisted officer, <b>Intake first</b> , Awarded the best cadet officer at the military academy, Sri Lanka Army.                                                                                                           |
| 2011-2013 | <b>Trainee consultant</b> , Industrial services bureau (ISB), Sri Lanka.                                                                                                                                                                          |
| 2011      | <b>Research intern</b> - CIC Holdings, Sri Lanka.                                                                                                                                                                                                 |
| 2010      | <b>Industrial trainee</b> - Prima group of companies, Sri Lanka.                                                                                                                                                                                  |

## Key Skills

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|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Academic   | Formal teaching of undergraduate and graduate students: Graph neural networks, Computational systems biology, Network biology, scRNAseq data analysis, spatial transcriptomics data analysis, systems medicine, systems pharmacology.   |
| Industrial | Biomedical engineering, Computational biology, cheminformatics, biological modeling, biotech and biopharma, PLC, PVS & PIC systems, cell line development, upstream and downstream process in biomanufacturing, biomedical engineering. |
| Coding     | Python, R, Js, C++ ,HTML, MATLAB, Wolfram, Linux, PyTorch, $\text{\LaTeX}$ .                                                                                                                                                            |

## Professional Affiliations

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|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023-     | Department of chemical engineering, University of Arkansas, Fayetteville, AR 27701, USA.                                                                                                                 |
| 2022-2023 | Institute of biochemistry, molecular biology and biotechnology, University of Colombo, Colombo, Sri Lanka.                                                                                               |
| 2016-2023 | The key laboratory of aquaculture nutrition and feeds (Ministry of agriculture and rural affairs); the key laboratory of mariculture (Ministry of education), Ocean University of China, Qingdao, China. |
| 2012-2016 | Department of food science and technology, postgraduate institute of agriculture, University of Peradeniya, Sri Lanka.                                                                                   |
| 2007-2011 | Department of animal science, Uva Wellassa University, Badulla, Sri Lanka.                                                                                                                               |

## List of Awards, Honors, and Outstanding Achievements

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|-----------|------------------------------------------------------------------------------------------------------|
| 2022      | Outstanding researcher: Omics data science, Wixsee, Sri Lanka.                                       |
| 2021      | Award: Modern project proposal: Biotech and bioengineering, Shandong province, China.                |
| 2021      | Grant proposal selection, Innovation and entrepreneurial funding, Qingdao, China.                    |
| 2021      | Award: Foreign expert consultant, Category: Biomedical data science, Qingdao, China.                 |
| 2021      | Consultancy contract for 2 years, Yantai city, Yantai, China.                                        |
| 2020      | Award for the "Honor of excellent international graduate of 2020", Ocean University of China, China. |
| 2020      | Award: College first class Ph.D., Ocean University of China, China.                                  |
| 2019      | Award: Innovative startup: Wixsee Genomics, NW Provincial council, Sri Lanka.                        |
| 2018      | Innovative research, Elsevier international conference, Qingdao, China.                              |
| 2016-2020 | Full scholarship by chinese scholarship council (CSC)- For Ph.D., Government, China.                 |
| 2014      | The honor of best officer Cadet, DE II-2014, Military academy, Sri Lanka.                            |
| 2011      | Second class upper honors, Uva Wellassa University, Sri Lanka.                                       |
| 2007-2011 | Full scholarship for bachelor's degree, Government, Sri Lanka.                                       |
| 2001-2003 | "Nana Nuwana" full scholarship for the best performance in GCE O/L, Government, Sri Lanka.           |
| 2000      | Award: Best performance, Sri Sumangala College, Sri Lanka.                                           |
| 1999      | Award: Best performance, Sri Sumangala College, Sri Lanka.                                           |

## Key Certificates

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| 2023-2025 | Certificates of graduate teaching. Subjects: scRNAseq data analysis, Cell culture systems. (3 certificates for 3 consecutive years)                                                                                  |
| 2023      | Certificate of appreciation: Lecturer computational systems biology, Biomedical data science, University of Arkansas, USA.                                                                                           |
| 2023      | Certificate: Hands-on start to wolfram mathematica, WOLFRAM U.                                                                                                                                                       |
| 2023      | Training certificates: Chemical hygiene plan training, Principles of biosafety, Compressed gas safety, Fire safety and fire extinguishers, Laboratory safety, Hazardous waste training, University of Arkansas, USA. |
| 2022      | Introduction to genomic technologies, Johns Hopkins University, USA.                                                                                                                                                 |
| 2021      | Machine learning for biomedical data science and drug discovery, Pine.bio, USA.                                                                                                                                      |
| 2021      | Breast cancer prediction using machine learning, Coursera project.                                                                                                                                                   |
| 2021      | Reverse and complement nucleic acid sequences (DNA, RNA) using python, Coursera project.                                                                                                                             |
| 2005      | Diploma in English, National youth service council, Sri Lanka.                                                                                                                                                       |
| 2004      | Computer certificate course, UNESCO Project, Sri Lanka.                                                                                                                                                              |
| 2001-2003 | Diploma in computer: Accounts and data base management, IRET computer institute, Sri Lanka.                                                                                                                          |

## Participation and Contributions

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|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025-June | Seminar (online)- Intelligence amplified: Human-centered education in the age of AI.                                                                    |
| 2024-2025 | Senior lecturer (online)- Cell systems modeling, IBMBB, University of Colombo, Sri Lanka.                                                               |
| 2023      | Lecturer: Systems biology, Biomedical data science, University of Arkansas, USA.                                                                        |
| 2023      | Senior lecturer: scRNAseq data analysis, Systems biology, IBMBB, University of Colombo, Sri Lanka.                                                      |
| 2022      | scRNAseq Data analysis: Illumina Nextseq 1000 data, Cancer hospital, Sri Lanka.                                                                         |
| 2022      | Lung disease research scRNAseq data analysis, University of Arizona, USA.                                                                               |
| 2022      | Certificate on precision oncology: Determinants of CAR T-cell clinical response in lymphoma and glioma, precision oncology news, powered by Genome web. |
| 2021      | Guest invitee: Key foreign outsourcing high-tech leaders conference, Yantai, Shangdong, China.                                                          |
| 2021      | Chief biotech consultant: Lysine production plant, Oryzas Organics (PVT) Ltd., Sri Lanka.                                                               |
| 2018      | Founder of Wixsee (startup in biomedical data science and precision medicine), Kurunegala, Sri Lanka.                                                   |

## Grants and Funding

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| 2023-2026 | National Institutes of Health (NIH) (NIGMS R01GM157589) and the Department of Defense (DOD) (DEPSCoR FA9550-22-1-0379).                                                                                                                                                        |
| 2023-2026 | FD201, Federal grants and contracts, FNO220 Individual and project research, GR017435 UAF, Inferring network models from spatial snapshots Richardson ORIG award, Grant: GR017435 UAF.                                                                                         |
| 2023      | Key R&D program of Shandong Province, China (2021SFGC0701).                                                                                                                                                                                                                    |
| 2020      | National Key R&D Program of China (2019YFD0900200) and the Key R&D Program of Shandong Province, China (2016CYJS04A01, 2017CXGC0105).                                                                                                                                          |
| 2019      | National natural science foundation (NSF) of China (No. 31572628), the Marine economic innovation and development regional model city project (2016) of Qingdao, and the fundamental research funds for the central universities of Ocean University of China (No. 201562017). |
| 2019      | National natural science foundation (NSF) of China (Grant no. 31572628) and Key R&D program of Shandong Province, China (Grant no. 2017CXGC0105).                                                                                                                              |

## Conference Presentations, Invited Talks & Posters

- 2026 Microsoft AI summit, Houston, TX, USA (in-person).
- 2024 Talk: NWA tech summit, Rogers, AR, USA (in-person).
- 2021 Talk: International biotech expo (held with Covid-19 safety guidelines), Nanning, China (in-person).
- 2018 Poster and presentation: Elsevier - 3<sup>rd</sup> International aquaculture conference. Title: Effects of dietary taurine and carbohydrates on muscle growth of turbot *Scophthalmus maximus* (In-person).

## Publications

### Manuscripts under review:

1. Hays H, Richardson WJ. Quantum computing enables global parameter estimation in ODE-based cell signaling networks. *Cell Syst* 2026.
2. Hays H, Richardson WJ. Path-dependent dynamics and asymmetric relaxation in cardiac fibroblast signaling define a mechanistic therapeutic window. *Syst Biol* 2026.

### Pre-prints:

1. Hays H. Hebbian-Oscillatory Co-Learning. *ArXiv* 2026. <https://doi.org/10.48550/arXiv.2603.08731>
2. Hays H. Selective Synchronization Attention. *ArXiv* 2026. <https://doi.org/10.48550/arXiv.2602.14445> **GitHub:** OSN
3. Hays H, Richardson WJ. A Survey of LLMs in Drug Discovery and Precision Medicine. *SSRN, ResearchGate* 2026. <https://doi.org/10.13140/RG.2.2.18610.47045>
4. Hays H. Resonant Sparse Geometry Networks. *ArXiv* 2026. <https://doi.org/10.48550/arXiv.2601.18064> **GitHub:** RSGN
5. Hays H. Encyclopedia of Large Language Models and Foundation Models. *Book, Zenodo* 2026. <https://doi.org/10.5281/zenodo.18261143>
6. Hays H. Attention mechanisms in neural networks. *ArXiv* 2026. <https://doi.org/10.48550/arXiv.2601.03329>
7. Hays H. CEFT: Consciousness-entropy field theory. *Authorea* 2025. <https://doi.org/10.22541/au.174231281.17403092/v2>
8. Hays H, Zhang W, Mai K. Synergistic effects of dietary taurine and carbohydrates supplementation on skeleton muscle of juvenile turbot *Scophthalmus maximus*. *bioRxiv* 2023. <https://doi.org/10.1101/2023.05.10.540249>

### Manuscripts submitted (Pre-prints available):

1. Hays H, Richardson WJ. A Survey of LLMs in Drug Discovery and Precision Medicine. *Frontiers in Drug Discovery, section In silico Methods and Artificial Intelligence for Drug Discovery*(Accepted- Submission number: 1854899) 2026. <https://doi.org/10.48550/arXiv.2602.00586>
2. Hays H, Richardson WJ. Hierarchical Molecular Language Models (HMLMs). *Nat Comm*(Submission number: NCOMMS-25-101042) 2025. <https://doi.org/10.48550/arXiv.2512.00696> **GitHub:** HMLMs

### Articles in international journals:

1. Hays H, Richardson WJ. RAG-GNN: retrieval-augmented graph neural networks for protein interaction network embeddings. *Front. Artif. Intell* 2026. <https://doi.org/10.3389/frai.2026.1851917> **GitHub:** RAG-GNN
2. Hays H, Richardson WJ. ECMSim: A high-performance interactive web application for real-time spatiotemporal simulation of cardiac ECM signaling and diffusion. *Matrix Biol Plus* 2026. <https://doi.org/10.1016/j.mbpplus.2026.100195> **GitHub:** ECMSim
3. Arif M, He M, Sameen NA, Hays H, Wang L, Zhang X, Zhang B, Li G, Yu T. Targeted catalytic eradication of antibiotic-resistant *Helicobacter pylori* by a pH-activated Fe-doped nanozyme. *J. Colloid Interface Sci.* 2026. <https://doi.org/10.1016/j.jcis.2026.140494>
4. Perera DN, Palliyaguruge CL, Eapasinghe DD, Karunathilake KG, Samarakoon SA, Perera PS, Hays H, et al. Factors affecting iron absorption and the role of fortification in enhancing iron levels. *Nutr Bull* 2023. <https://doi.org/10.1111/nbu.12643>
5. Hays H, Gu Z, Mai K, Zhang W. Transcriptome-based nutrigenomics analysis reveals the roles of dietary taurine in the muscle growth of juvenile turbot (*Scophthalmus maximus*). *Comp Biochem Physiol D* 2023. <https://doi.org/10.1016/j.cbd.2023.101120>
6. Sampath WWHA, Rathnayake RMDS, Yang M, Zhang W, Mai K. Roles of dietary taurine in fish nutrition. *Mar Life Sci Technol* 2020;2:360-375. <https://doi.org/10.1007/s42995-020-00051-1>
7. Sampath WWHA, Zhang Y, Liu J, Yang M, Zhang W, Mai K. Dietary taurine improves muscle growth and texture characteristics in juvenile turbot (*Scophthalmus maximus*). *Aquac Rep* 2020;17:100305. <https://doi.org/10.1016/j.aqrep.2020.100305>
8. Yang M, Deng K, Pan M, Zhang Y, Sampath WWHA, Zhang W, Mai K. Molecular adaptations of glucose and lipid metabolism to different levels of dietary carbohydrates in juvenile Japanese flounder *Paralichthys olivaceus*. *Aquac Nutr* 2020;26:516-527. <https://doi.org/10.1111/anu.13013>
9. Liu D, Deng K, Sampath WWHA, Gu Z, Pan M, Zhang Y, Zhang W, Mai K. Responses of glucosensing system to glucose in Japanese flounder *Paralichthys olivaceus* fed diets with different carbohydrate content. *Comp Biochem Physiol B* 2019;232:72-78. <https://doi.org/10.1016/j.cbpb.2019.03.001>
10. Sampath WWHA, Zhang Y, Gu Z, Liu J, Zhang W, Mai K. Effects of taurine and carbohydrates on muscle growth of turbot (*Scophthalmus maximus*). Abstract. 3rd International Conference of *Aquaculture* 2018:25-28. <https://doi.org/10.1101/2023.05.10.540249>
11. Sampath WWHA, Nambapana M, Ganegoda GAP. Study on the effect of curry leaves supplementation of broiler feed on feed conversion ratio. *J Agric Environ Sci* 2013;ISSN:2304-1234. <https://erepo.lib.uwu.ac.lk/items/3e79334b-d90b-45f3-82f1-32d8c2b74ffb>